

Original Article

Obstructive sleep apnea during REM sleep: effects on morning and evening blood pressure

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Abstract

Study Objectives: Obstructive sleep apnea (OSA) is linked to the emergence and progression of cardiovascular complications including hypertension, stroke, arrhythmias, coronary artery disease, and heart failure. Epidemiological studies have reported that hypertension is associated with respiratory events during REM sleep. We examined the relationship between respiratory events during REM and morning and evening hypertensive blood pressure (BP) levels in a clinical sleep population.

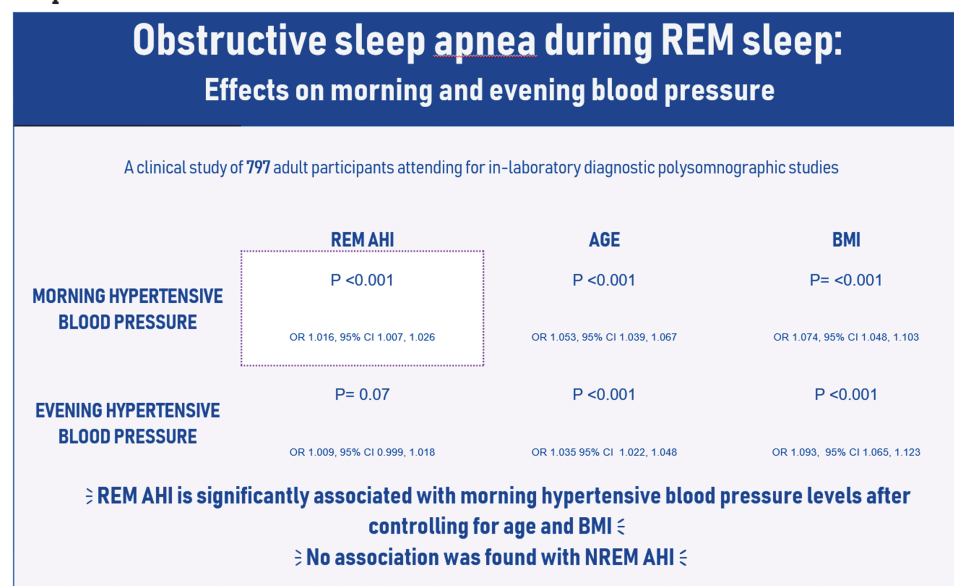
Methods: This study included data from in-laboratory diagnostic polysomnographic studies ($n = 797$) from adults attending for investigation of OSA. Hypertensive BP levels were defined using BP measurements taken in the evening before and morning after polysomnography, and the use of antihypertensive medication. Regression modeling was undertaken to examine the probability of evening and morning hypertensive BP levels according to REM apnea-hypopnea index (AHI), NREM AHI, gender, age, body mass index (BMI), alcohol use, total sleep time (TST), sleep time $SpO_2 < 90\%$, and smoking status.

Results: The probability of morning hypertensive BP levels was significantly independently associated with age ($p < .001$), BMI ($p < .001$), and REM AHI ($p < .001$). No significant effect was found for the male gender, NREM AHI, alcohol use, TST, sleep time $SpO_2 < 90\%$, or smoking ($p > .05$ for all). The probability of evening hypertensive BP levels was only significantly associated with age ($p < .001$), male gender ($p = .012$), BMI ($p < .001$), and TST ($p = .032$).

Conclusions: Respiratory events during REM sleep are significantly associated with morning hypertensive BP levels. Future research is needed to determine whether treatment of these events can prevent or reverse morning hypertensive BP levels.

Key words: obstructive sleep apnea; sleep-disordered breathing; rapid eye movement; blood pressure; hypertension

Graphical Abstract



Statement of Significance

This study reports the risk of evening and morning hypertensive blood pressure levels according to REM AHI in a clinical population of patients referred for suspected OSA. Morning hypertensive blood pressure levels were independently associated with REM AHI, not NREM AHI, after controlling for age, and BMI. This study provides clinical context in a cohort of men and women over a wide age range to support and extend the findings of epidemiological studies that explored the relationship between REM-OSA and hypertension. When evaluating the effects of sleep-disordered breathing, BP measurement in the morning may provide a better assessment of cardiovascular risk.

Introduction

Obstructive sleep apnea (OSA) is a highly prevalent condition characterized by repetitive events of complete (apnea) or partial (hypopnea) collapse of the upper airway during sleep [1–3]. OSA causes intermittent hypoxemia and hypercapnia, sleep fragmentation, and cardio-sympathetic changes such as increased blood pressure and heart rate [1, 2, 4]. These acute physiological changes are associated with adverse health outcomes, including increased incidence of cardiovascular, and metabolic disease [1–5].

OSA is commonly observed during rapid eye movement (REM) sleep and may even occur exclusively within this sleep state [3, 6]. REM-OSA is more commonly observed in women, and in patients with mild to moderate OSA [5, 7]. Compared to Non-REM (NREM) sleep, respiratory events in REM tend to be more frequent, longer in duration, and are more likely to be associated with hypoxemia [3, 6, 7]. The increased severity of OSA in REM can be attributed to reduced genioglossus muscle activity during REM, which leads to increased muscle collapsibility and obstruction in the upper airway [3, 5, 7]. Increased duration of respiratory events and more significant oxygen desaturations may also be explained by decreased hypercapnic and hypoxic ventilatory drive during REM [3].

OSA is linked to the emergence and progression of cardiovascular disease including hypertension, stroke, arrhythmias, coronary artery disease, and heart failure [1, 8]. REM sleep in particular is associated with cardiovascular instability and increased sympathetic activity [3, 5], which may suggest that REM-based OSA could precipitate cardiovascular risk more significantly than NREM-related OSA [3, 9].

Whilst there have been numerous studies detailing the association between OSA and hypertension, recent epidemiological studies have found strong associations between REM-OSA and hypertension which may uniquely increase cardiovascular risk [3–5]. Mokhlesi et al. demonstrated a dose-responsive relationship between hypertension and REM-OSA in the Wisconsin Sleep Cohort [3] utilizing polysomnography (PSG) and evening measurements of blood pressure. Importantly, this study was the first to determine a significant association between REM-OSA and hypertension, independent of NREM-OSA, and controlled for known confounding factors including age, gender, and smoking status [3]. Similarly, the longitudinal study of the Men Androgens Inflammation Lifestyle Environment and Stress (MAILES) cohort investigated hypertension and undiagnosed OSA [10]. This study also supported an independent association between REM-OSA and hypertension in men >35 years of age using a clinical measurement of blood pressure [10]. Conversely, a study of the population-based HypnoLaus cohort found that REM-related sleep-disordered breathing (REM-SDB) was associated with metabolic syndrome and diabetes, known risk factors for cardiovascular disease [1, 5]; however, a significant relationship between hypertension and REM-SDB was only found with a REM AHI >20 events/h [5].

To supplement the findings of these studies, more research is required to circumstantiate the role of REM-based OSA as an antecedent of hypertension. As hypertension could be a critical mediator between OSA and cardiovascular risk, it is important to determine how significantly REM-OSA can influence or predict hypertension [3, 4]. Till date, this has not been explored in a clinical population referred for investigation of OSA, nor utilizing both evening and morning measures of blood pressure. Accordingly, the aim of the current study was to investigate the role of REM-OSA on the prevalence of hypertensive BP levels in a clinical sleep cohort. Specifically, this study explored the associations between OSA during REM and NREM sleep and evening and morning blood pressure to test the hypothesis that both evening and morning hypertensive BP levels are associated with REM AHI.

Methods

Participants

Data from all in-laboratory diagnostic PSG studies for adults attending for investigation of OSA performed at Eastern Health (Melbourne, Australia) over a 16-month period were reviewed ($n = 1080$). No split night sleep studies were included and all patients were naïve to PAP therapy. Patients with less than 3 hr of TST or less than 30 min of REM sleep were excluded, consistent with previous studies (Figure 1). The final sample size for analysis was 797. All participants had evening blood pressure measured (PM) and 781 also had morning blood pressure measured (AM). Ethical approval for this study was obtained by the Eastern Health Human Research Ethics Committee. The data

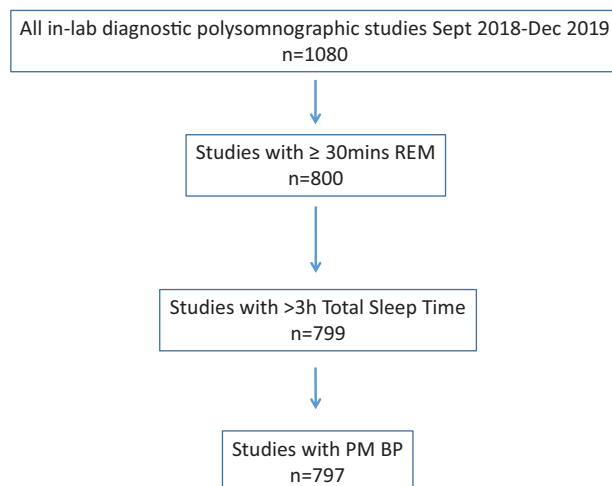


Figure 1. Flowchart of study data inclusion.

underlying this article cannot be shared publicly due to the privacy of individuals that participated in the study. The data may be shared on reasonable request to the corresponding author.

Polysomnography

All participants underwent routine overnight PSG using a commercially available PSG system (Compumedics Grael). Electroencephalograms (EEG), Electrooculograms (EOG), chin Electromyogram (EMG), Electrocardiogram (ECG), left and right leg EMG and body position were recorded. Oxygen saturation (SpO_2) was measured by pulse oximetry, thoracic and abdominal breathing movements were recorded via uncalibrated respiratory inductance plethysmography, and airflow was recorded via nasal pressure and oro-nasal thermocouple. Sleep and arousal were scored from the EEG, EOG, and chin EMG channels in 30-second epochs according to standard criteria. Obstructive apneas, mixed apneas, and hypopneas were defined according to the American Academy of Sleep Medicine standard [11]. Hypopneas were scored based on the recommended criteria of a reduction in flow of $\geq 30\%$ for at least 10 s with an associated oxygen desaturation of $\geq 3\%$ or an arousal from sleep.

Prior to PSG, height, and weight were recorded, and BMI was calculated. Waist and hip circumferences were measured, and the waist-hip ratio was calculated. Waist circumference was measured at the midpoint between the lower margin of the last palpable rib and the top of the iliac crest. Hip circumference is measured around the widest portion of the buttocks. The Epworth Sleepiness Scale (ESS) was recorded. Prior to the polysomnography patients completed standardized questionnaires documenting smoking status, alcohol intake, past medical history, and current medications. Alcohol intake was assessed using the *Alcohol Use Disorders Identification Test* (AUDIT) [12], a 10-item instrument developed by the World Health Organization to screen for alcohol use disorders.

Blood pressure

Systolic and diastolic blood pressure was measured before PSG in the evening and after PSG in the morning using a standard sphygmomanometer (Philips Intellivue). Evening BP measures were taken between 07:00 and 08:00 pm prior to the overnight sleep study. Morning BP measurements were taken between 06:00 and 07:00 am after the sleep study. Measurement was taken with the patient seated in a comfortable position, with arm flexed and elbow at the level of the heart, with the cuff placed on the upper arm so that its bottom edge is 2–3 cm above the antecubital fossa. Hypertensive BP levels were defined as either BP $\geq 140/90$ mmHg or the current use of antihypertensive medication based on patient self-report (a-adrenergic antagonists, b-blockers, calcium-channel blockers, diuretics, angiotensin-converting enzyme inhibitors, or angiotensin-receptor blockers).

Statistics

Demographics and polysomnographic variables were compared across REM AHI intervals (approximate quartiles; REM AHI <5 events/h, 5–<25 events/h, 25–<50 events/h, ≥ 50 events/h: Table 1) using ANOVA on ranks with Dunn's post hoc analysis as data failed normality testing. Data are presented as median (IQR) with significance taken at the $p < .05$ level. Differences in gender distribution and %PM and %AM hypertensive BP levels across REM AHI intervals were analyzed with χ^2 analysis. Analysis was performed using Sigmaplot version 12.3 statistical software.

For regression modeling data were analyzed with R version 4.1 [13]. A p -value of $< .05$ was considered significant. Multivariable logistic regression models were fitted with AM hypertensive BP levels (respectively, PM hypertensive BP levels) as the response. Explanatory variables thought from previous knowledge to be relevant were taken as candidates for inclusion in the models. Those which did not have significant coefficients (and associated odds ratios) in the model were generally omitted, in the interests of parsimony. The exception was in the case of the variables shown in Tables 2 and 3, which show the results from multivariable logistic regression models leading to Figures 2 and 3. We assessed for collinearity between variables entered into the models and confirmed that variance inflation factors were in the range 1–2, and none were higher than three suggesting no significant collinearity. Plots of logit against REM AHI showed saturation of the effect for this variable (values above 65 were all effectively the same), so the variable was transformed so that REM AHIs above 65 were now equal to 65. Similar transformations of the NREM AHI made no effective difference. These models and figures show the estimated probabilities after adjusting for the indicated variables, whether significant or not. Overall significance of the fitted models was measured by likelihood ratio tests of the fitted models against the model with just an intercept.

Results

Within the final cohort ($n = 797$), 57% of participants were male and the age ranged from 18 to 87 years. The proportions of patients diagnosed with no OSA (Total AHI < 5 events/h), mild OSA (Total AHI 5–<15 events/h), moderate OSA (Total AHI 15–<30 events/h) and severe OSA (Total AHI ≥ 30 events/h) were 18%, 27%, 23%, and 32% respectively.

Demographics and polysomnographic variables of the 797 participants are presented across intervals of REM AHI in Table 1. As expected, both total AHI and NREM AHI increased with increasing REM AHI. Furthermore, ODI 3%, REM ODI 3%, time spent with $\text{SpO}_2 < 90\%$ and arousal index increased with increasing REM AHI. In addition, BMI and WHR also increased across intervals. Those with REM AHI <5 events/h were significantly younger than the other three intervals. There was no difference in gender distribution or ESS across the groups. The REM AHI ≥ 50 events/h group consistently had higher BP (both systolic and diastolic) compared to other groups ($p < .05$). 461/797 (58%) had PM hypertensive BP levels. 405/781 (52%) had AM hypertensive BP levels. Proportions of patients with PM and AM hypertensive BP levels both significantly increased with increasing REM AHI across the intervals ($p < .05$ for both). The number of patients taking antihypertensive medication was 253 (31.7%).

Evening hypertensive BP levels

Results from the multivariable logistic regression model estimating the probability of evening hypertensive BP levels are summarized in Table 2. The evening hypertensive BP levels model included a sample of 771 patients with all variables recorded (from the total sample 26 did not have Audit alcohol scores). The probability of evening hypertensive BP levels was significantly associated with age ($p < .001$), male gender ($p = .012$), BMI ($p < .001$), and TST ($p = .032$). No significant effect was found for, alcohol use, sleep time $\text{SpO}_2 < 90\%$ (T90%), or smoking ($p > .05$ for all). Similarly, there was no significant effect for evening hypertensive BP levels based on REM AHI ($p = .07$) or NREM AHI ($p = .9$).

Table 1. Patient demographics and polysomnographic variables according to REM AHI interval

	REM AHI < 5/h	REM AHI 5–<25/h	REM AHI 25–50/h	REM AHI ≥ 50/h	p value
n	160	259	190	188	
Age (y)	44.5 (32–56)*	52 (44–63)	54 (45–66)	53 (43–60)	<.001
ESS	9 (5–13)	8 (5–12)	8 (4–12)	9 (5–12)	n.s.
% Male	56	59	56	56	n.s.
BMI (kg/m ²)	27.9 (24.6–31.6)*	29.4 (26.7–34.0)*	32 (27.4–36.3)*	37.5 (30.5–45.1)*	<.001
WHR	0.91 (0.86–0.97)*	0.95 (0.90–0.99)*	0.95 (0.90–1.00)	0.96 (0.91–1.03)	<.001
REM AHI (/h)	2.2 (1.0–3.4)*	12.1 (8.5–17.7)*	36.8 (31.5–43.7)*	64.8 (57.7–78.3)*	<.001
NREM AHI (/h)	3.6 (1.6–11.5)*	12.0 (4.5–21.2)*	18.1 (8.7–34.9)*	46.5 (25.1–71.8)*	<.001
Total AHI (/h)	3.4 (1.6–9.8)*	12.2 (6.4–20.2)*	21.4 (13.1–35.5)*	49.6 (30.6–71.7)*	<.001
T90 (%)	0.0 (0.0–0.1)*	0.2 (0.0–1.2)*	1.3 (0.3–4.5)*	8.6 (3.0–27.9)*	<.001
ODI 3% (/h)	1.5 (0.4–5.3)*	6.8 (3.4–15.6)*	16.6 (8.9–30.5)*	42.4 (25.2–68.9)*	<.001
REM ODI 3% (/h)	0.7 (0.0–2.3)*	9.3 (4.7–14.6)*	31.5 (22.9–40.4)*	61.9 (53.1–73.9)*	<.001
Arousal Index (/h)	11.6 (8.3–18.9)*	15.9 (10.1–23.5)*	17.7 (12.7–27.8)*	25.9 (16.4–46.7)*	<.001
PM BP (mmHg)					
- Systolic	129 (119–141)*	131 (120–143)	133 (123–147)	140 (128–154)*	<.001
- Diastolic	77 (68–86)	78 (71–86)	79 (71–86)	83 (75–93)*	<.001
AM BP (mmHg)					
- Systolic	121 (111–131)*	127 (119–142)	130 (119–143)	138 (126–150)*	<.001
- Diastolic	74 (66–83)*	77 (68–87)	79 (68–89)	86 (74–93)*	<.001
% PM hypertensive BP levels	42	52	60	76	<.001
% AM hypertensive BP levels	29	47	53	73	<.001

Data presented as median (IQR).

*p < .05 compared to all other groups in post hoc analysis.

^p < .05 compared to REM AHI 25–<50 events/h in post hoc analysis.

ESS: Epworth Sleepiness Score; BMI: body mass index; WHR: waist-hip ratio; T90: sleep time SpO₂ <90%;

ODI: oxygen desaturation index.

AM hypertensive BP levels sample size 781.

Table 2. Multivariable logistic regression model estimating the probability of evening hypertensive BP levels

Characteristic	OR ¹	95% CI ¹	p-value
Age (y)	1.035	1.022, 1.048	<.001
Gender			
F	—	—	
M	1.561	1.103, 2.216	.012
BMI (kg/m ²)	1.093	1.065, 1.123	<.001
REM AHI (events/h)	1.009	0.999, 1.018	.07
NREM AHI (events/h)	1.000	0.99, 1.010	.9
Audit alcohol score	1.012	0.97, 1.054	.5
Total sleep time (mins)	0.997	0.99, 1.000	.032
T90% (sqrt %Time SpO ₂ <90)	1.011	0.995, 1.031	.2
Smoker			
N	—	—	
Y	0.840	0.523, 1.348	.5

¹OR = Odds Ratio, CI = Confidence Interval.

Null deviance = 1,051; Null df = 770; Deviance = 889; Residual df = 761; No. Obs. = 771.

Likelihood ratio test of model adequacy had a p-value < .001. In a subset analysis of patients that were not on antihypertensive medication (n = 544), similar results were found with again no

significant effect for evening hypertensive BP levels based on REM AHI (p = .09) or NREM AHI (p = .89).

Figure 2 models the change in the estimated probability of hypertensive BP levels according to REM AHI across age groups (20, 40, 60, and 80 years) and BMI categories (range 20–80 kg/m²). Whilst the probability of evening hypertensive BP levels increased with REM AHI, this failed to reach statistical significance.

Morning hypertensive BP levels

Results from the multivariable logistic regression model estimating the probability of morning hypertensive BP levels are summarized in Table 3. The morning hypertensive BP levels model included a sample of 755 patients with all variables recorded (from total sample 26 did not have Audit alcohol scores and a further 16 did not have morning BP measurements). The probability of morning hypertensive BP levels was significantly associated with age (p < .001), BMI (p < .001), and REM AHI (p < .001). No significant effect was found for the male gender, NREM AHI, alcohol use, TST, sleep time SpO₂ <90% (T90%), or smoking (p > .05 for all). Likelihood ratio test of model adequacy had a p-value < .001. In a subset analysis of patients that were not on antihypertensive BP levels medication (n = 531), similar results were found with a significant effect for morning hypertensive BP levels based on REM AHI (p < .001), but again no effect of NREM AHI (p = .92).

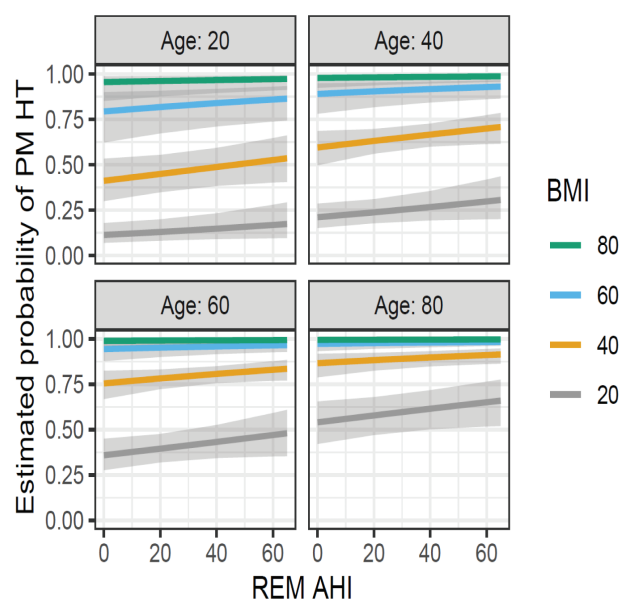
Figure 3 models the change in estimated probability of hypertensive BP levels according to REM AHI across age groups (20, 40,

Table 3. Multivariable logistic regression model estimating the probability of morning hypertensive BP levels

Characteristic	OR ¹	95% CI ¹	p-value
Age (y)	1.053	1.039, 1.067	<.001
Gender			
F	—	—	
M	1.173	0.820, 1.681	.4
BMI (kg/m ²)	1.074	1.048, 1.103	<.001
REM AHI (events/h)	1.016	1.007, 1.026	<.001
NREM AHI (events/h)	1.002	0.99, 1.011	.8
Audit alcohol score	1.008	0.97, 1.050	.7
Total sleep time (mins)	0.998	0.99, 1.001	.1
T90% (sqrt %Time SpO ₂ <90)	1.003	0.99, 1.019	.7
Smoker			
N	—	—	
Y	0.750	0.459, 1.218	.2

¹OR = Odds Ratio; CI = Confidence Interval.

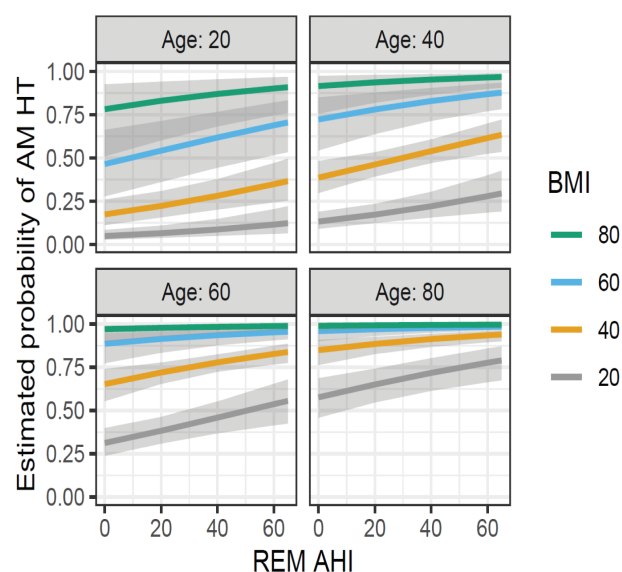
Null deviance = 1,046; Null df = 754; Deviance = 849; Residual df = 745; No. Obs. = 755.

**Figure 2.** Estimated probability of evening hypertensive BP levels (PM HT) with 95% confidence bands (shaded areas) according to REM AHI at 4 different ages and 4 BMI values. Estimates were obtained by logistic regression, after adjusting for the significant terms: Age, gender, BMI, TST (total sleep time); and the nonsignificant terms: REM AHI, NREM AHI, Audit, T90, and smoking status.

60, and 80 years) and BMI categories (range 20–80 kg/m²). Higher REM AHI was associated with an increased probability of morning hypertensive BP levels increasing in a dose-responsive fashion according to age and BMI.

Discussion

This study, for the first time, reports the risk of evening and morning hypertensive BP levels according to REM AHI in a clinical population of patients referred for suspected OSA. Using both evening and morning blood pressure measurements in conjunction with

**Figure 3.** Estimated probability of morning hypertensive BP levels (AM HT) with 95% confidence bands (shaded areas) according to REM AHI at four different ages and four BMI values. Estimates were obtained by logistic regression, after adjusting for the significant terms: Age, BMI, REM AHI; and the nonsignificant terms: gender, NREM AHI, Audit, TST (total sleep time), T90, and smoking status.

medication history to define hypertensive BP levels before and after polysomnography, the study allows for a novel perspective on the relationship between REM-OSA and hypertensive BP levels. Our study demonstrated that morning hypertensive BP levels were independently associated with REM AHI, not NREM AHI, after controlling for age and BMI. This study provides clinical context in a cohort of men and women over a wide age range to support and extend the findings of epidemiological studies that explored the relationship between REM-OSA and hypertension.

These findings support and extend those reported in large epidemiological studies in recent years [3, 5, 10]. Large epidemiological studies in non-referred cohorts have demonstrated a significant association between REM AHI and hypertension using a clinic measurement of BP. This study however now provides further evidence for this association in a real-world clinical cohort of patients referred for investigation of OSA, using both evening and morning blood pressure measures across a range of individuals with larger REM AHIs. Almost 50% of the present cohort had a REM AHI ≥ 25 events/h, reflecting the acuity of patients typically referred to a sleep laboratory for investigation of sleep-disordered breathing. The benefit of assessing this in a clinical population is the transferability of results to real-world situations, particularly as it has recently been shown that even the findings of OSA treatment-related randomized controlled trials may not be generalizable to sleep clinic OSA patients as the populations differ significantly [14]. The additional effects of age and BMI cannot be discounted in this cohort. Prevalence of OSA increases with increasing age and weight [15], and untangling the negative cardiovascular effects of these can be challenging. For this reason, we have estimated the risk of hypertensive BP levels according to AHI across four ages and four BMIs to elucidate the combined effects. In extending our analyses, we also analyzed REM AHI associations with hypertensive BP levels in patients with NREM AHI <5 events/h ($n = 180$). In the models, age and BMI remain significant but neither REM nor NREM were independently associated with BP, likely due to the small sample size (data not shown).

In this study, REM AHI was shown to be associated with an increased probability of morning hypertensive BP levels, whilst no association was found with NREM AHI. With normal sleep architecture, there is a higher proportion of NREM sleep early in the night and a greater proportion of REM sleep later in the night. REM sleep periods tend to lengthen, with more REM often occurring close to the end of the sleep period. REM sleep is associated with reduced genioglossus single motor unit activity, increasing the propensity for upper airway collapse [7]. Additionally, REM has reduced hypoxic and hypercapnic ventilatory responses compared with NREM and wakefulness, effectively lengthening the time to arousal response [16, 17]. Importantly, respiratory events are known to be of longer duration in REM sleep and associated with more significant hypoxia [18]. Sustained sympathetic activation driven by chemoreflex activation from intermittent hypoxia and resaturation (long-term facilitation) is the most likely mediator of the increased cardiovascular risk seen in OSA [19–21]. Elevated sympathetic activity that has been postulated as a mechanism for systemic hypertension in OSA by peripheral vascular remodeling and increased vascular resistance. Therefore, patients with OSA are likely to experience more frequent and more severe obstructive events closer to waking time in the morning and are more likely to experience increased sympathetic activity and cardiovascular instability [3, 5] during the morning prior to waking. Accordingly, because REM was likely to have occurred in close proximity to the morning blood pressure measurement, this study may therefore indicate a temporal relationship between morning hypertensive BP levels and REM-AHI.

This study demonstrated a significant relationship between REM AHI and the probability of morning hypertensive BP levels, but not evening hypertensive BP levels. REM AHI just failed to reach significance in the multivariable logistic regression model, although the proportion of patients with evening hypertensive BP levels increased significantly across increasing REM AHI intervals. One reason as to why REM AHI was not significant for evening hypertensive BP levels may be the timing of antihypertensive medication. Whilst we did not record when patients were administered this medication, it may be that daytime dosing effectively reduced BP so that the evening measure was blunted. Second, the close temporal relationship between REM and the morning measurement but not the evening measurement may also have an effect. Evening hypertensive BP levels showed a significant correlation with both age and BMI. However, unlike morning hypertensive BP levels, there was also a significant relationship observed between evening hypertensive BP levels and decreased total sleep time (TST). Previous studies have linked short sleep duration to a higher risk of hypertension [22]. It is possible that TST in this study is reflective of a patient's typical sleep duration; however, it is unclear why this current study observed this relationship through only the evening blood pressure measurements.

Clinical implications

The relationship between REM AHI and hypertensive BP levels was shown to be significant in the morning, and this finding may have clinical implications for the use of CPAP therapy [3, 5]. Given that REM sleep tends to increase in frequency and duration in the latter half of the night, treatment of REM-SDB may therefore be impacted by the duration of CPAP use. Trends in CPAP use over 20 years have shown that poor adherence continues to present a significant challenge for the effective treatment of OSA [23], meaning that the associated cardiovascular risk factors including hypertension may also be poorly mitigated. As the benchmark

for CPAP adherence is typically 4 hr of use per night, removing treatment midway through the night may result in respiratory events during REM going untreated. Our study, as well as epidemiological studies in nonreferred cohorts, clearly demonstrate that events during REM play a critical role in the development of hypertensive BP levels. Leaving such events untreated may explain the lower-than-expected benefit of CPAP treatment on BP in randomized controlled trials [24, 25]. Furthermore, it is clear that improving CPAP adherence (i.e. likely using treatment longer during the night and therefore through REM later in the sleep period) increases the blood pressure-lowering effect [26, 27]. Grimaldi et al. was the first study to explore the relationship between hours of CPAP use and the adequate treatment of OSA in REM sleep [28]. In the context of glycemic control for subjects with type 2 diabetes, this study examined the mean cumulative minutes of NREM and REM sleep over 8 hr to demonstrate how different durations of optimal CPAP use could impact the effective treatment of OSA. CPAP adherence of approximately 4 hr per night resulted in 60% of REM sleep being left untreated, whilst 7 hr of CPAP use treated 87% of REM sleep [28]. Untreated events during REM towards the end of the night may have implications for the increased risk of sudden death from midnight to 06.00 am in patients not on CPAP [29] as well as an early morning wake-up ischemic stroke [30]. This risk is of considerable importance given that CPAP adherence is lower among patients with REM-only OSA compared to patients with nonstage specific OSA [31] and REM-related OSA is a reliable risk factor for dropout from CPAP therapy [32]. This study also has clinical implications for the timing of blood pressure measurement in OSA patients. Studies have often not stated the time of day of measurement, or only used daytime clinic measurements. As this study showed a significant association between REM AHI and morning hypertensive BP levels, any blood pressure measures in the population should be taken in the morning to better determine hypertensive risk.

Limitations

This study was conducted using a clinical cohort, with the diagnosis of sleep disorders being the primary goal. Measurements were taken from a large heterogeneous cohort, reflecting the real-world scenario of OSA in the community. As this was a clinical cohort, routine clinical BP measures were taken which involved a single BP measurement, which is in contrast to the epidemiological studies in the field which utilized two or three readings. Whilst the use of antihypertensive medication was used to define known hypertensive BP levels consistent with other studies on the topic, we did not record the timing of medication use. Given the study investigated evening and morning blood pressure measures, adding the timing of medication use as a co-variate may be beneficial in future studies. Morning measurements may have been confounded by the phenomenon of morning blood pressure surge, which is thought to occur upon rising [33, 34]. For this reason, continuous BP measurement throughout the polysomnography could be a superior measure. We did not measure acute BP changes during sleep which would have likely been associated with respiratory event duration [35] and arousal duration, and have been postulated to drive the long-term changes in blood pressure [36]. Additionally, the current study did not explore the effect of apnea and hypopnea duration on long-term BP changes. Future studies should consider assessing respiratory event duration when investigating cardiovascular outcomes of OSA as there is evidence it may be associated with chronic BP changes, not just acute changes [37].

Previous literature describes that patients with OSA have a rise in nighttime and morning BP, in effect losing their normal BP diurnal rhythm, compared with individuals without OSA. The reported evening-to-morning differences were small but nonetheless significant [38]. The raw BP measures in the current study do not show a morning rise in BP according to OSA severity. However statistical analysis revealed that there were differences between the groups—the REM AHI ≥ 50 events/h consistently had a higher BP (both systolic and diastolic) compared to other groups supporting the finding that REM AHI is associated with hypertensive BP levels. Importantly, the current study refers to this data as reflecting hypertensive BP levels, not hypertension, as hypertension is a disease characterized by a persistent elevation in BP, not a single reading.

Previous literature also describes an increased prevalence of REM-predominant OSA in women. Bahammam et al. reported that the prevalence of REM-predominant OSA in women was 25% compared with 12% in men using the following criteria: AHI of ≥ 5 /h, with REM-AHI/non-REM AHI of >2 , a non-REM AHI of <15 /h, and a minimum of 15 min of REM sleep [39]. The current study was not designed to explicitly look at REM-predominant OSA, but rather the REM AHI itself within a clinical population. Nonetheless, in our study 43% of participants in the final sample size were female, and this gender percentage was reflected across the REM AHI intervals ensuring that both genders were represented appropriately in the analysis. Finally, data were collected from a patient cohort at a single hospital site which may limit the transferability of the findings.

In conclusion, REM AHI is significantly associated with morning hypertensive BP levels after controlling for age and BMI. When evaluating the effects of sleep-disordered breathing, BP measurement in the morning may provide a better assessment of cardiovascular risk. Future research is needed to determine whether treatment of OSA during REM periods can prevent or reverse morning hypertensive BP levels.

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